Fall 2015 / EC571

Instructor:Ajay Joshi

Exam 1

Date: October 8, 2015

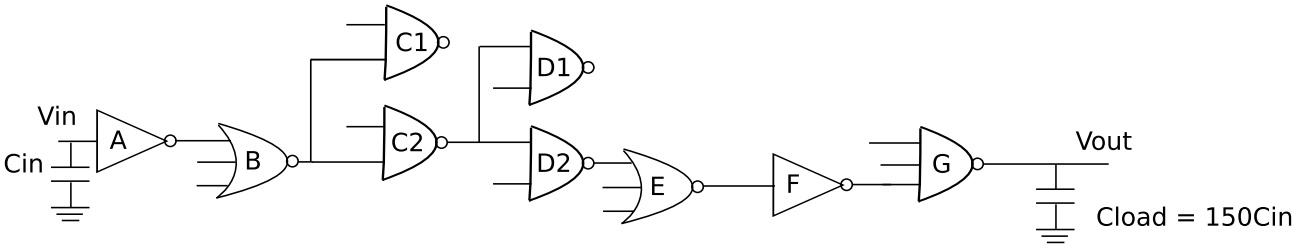
Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. The solutions should be written in the spaces provided on the test sheet.
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
7. Material sharing is not allowed.
8. Calculators are allowed.
9. Boston University Academic Code of Conduct applies.

Question 1



For the circuit shown in the above figure, using the logical effort technique, calculate the input width of transistors for each gate in the path such that total delay of the path is minimized. You can assume that gates C1 and C2 are identical and gates D1 and D2 are identical. Assume Cin = 4 fF, Cg = 1 fF/u. (20 points)

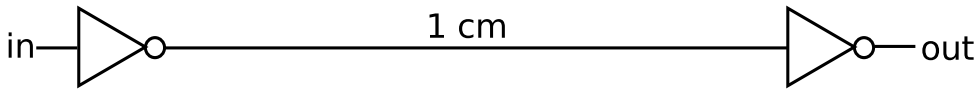
Question 2

1. Draw the static complementary CMOS logic implementation for the following function - $f(A,B,C) = \overline{A + B \cdot C}$. (6 points)

2. Determine the widths of the various transistors as a function of W so that worst case $\tau_{HL} = \tau_{LH}$ and the delay matches with the delay of an inverter. You can assume $2R_{sq}(NMOS) = R_{sq}(PMOS)$. Ignore the internal capacitances of the non-output nodes while determining the transistor sizing.(7 points)

3. Calculate τ_{HL} and τ_{LH} (Elmore delay) for both gates using the following parameters – width of the NMOS in the inverter W = 1.0 micron, L = 0.25 micron $R_{sq}(NMOS)$ = 10 Kohm/sq, $R_{sq}(PMOS)$ = 20 Kohm/sq, C_g = 1.2 fF/micron, C_d = 1.5 fF/micron, C_L = 100 fF. Include all internal capacitance. Do you get any mismatch between τ_{HL} and τ_{LH}? If yes, what can be done to match τ_{HL} and τ_{LH}? (7 points)

Question 3



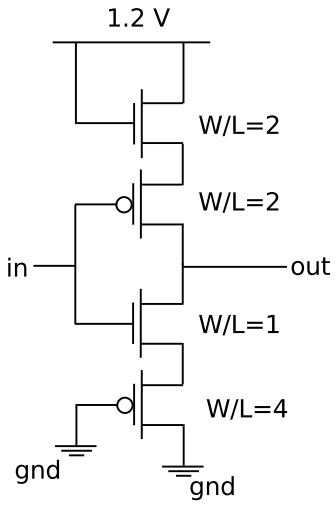
1. For the circuit shown in the above figure, draw the equivalent RC network that includes all the parts. Assume a lumped model for the interconnect and a load capacitance of C_L at the output. (4 points)

2. Write the Elmore delay equations for the entire circuit from in to out. (4 points)

3. Calculate the Elmore delay for the entire circuit. You can use the following technology parameters for calculating the delay - Vdd = 1.5 V, L = 0.25 micron Rsq(NMOS) = 10 Kohm/sq, Rsq(PMOS) = 25 Kohm/sq, Cg = 1 fF/micron, Cd = 1.5 fF/micron, wire capacitance = 0.25 fF/micron, wire resistance = 15 ohms/sq. The interconnect has a width of 0.5 micron. Assume that all inverters use a minimum width of 300 nm for the NMOS. You will need to choose the width of the PMOS transistor such that the low-to-high delay of the inverters matches with the high-to-low delay. Assume a load of 20 fF at the output. (4 points)

4. Your classmate suggests that it may be possible to reduce the delay of this circuit by inserting inverters at intermediate points along the wire. Verify your classmate's claim by determining the delay of the circuit for the case where you insert 2 inverters on the wire such that the wire gets divided into 3 equal parts with each part being driven by an inverter and there is an inverter at the end. You can assume the inserted inverters have the same size as that of the inverters at the two ends of the wire.(8 points)

Question 4



1. Using the simple switch model of the transistors with $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, $V_{th,N}$ = 0.2 V, $V_{th,P}$ = -0.2 V and V_{DD} = 1.2 V, draw the VTC of the circuit shown above. Provide the values for V_{OH}, V_{OL}, V_{IH} and V_{IL}. Explain your answer in detail. (15 points)

2. For the same circuit how much energy is supplied by the power supply V_{DD} to charge a load capacitor of 15 fF when we feed a step input ($V_{DD} \rightarrow 0$ V)? Ignore any charging or discharging of internal capacitances.

Question 5

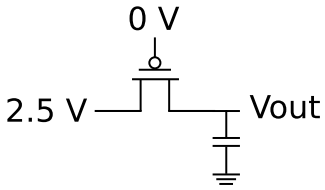
Design the static complementary CMOS implementation of an encoder circuit that takes a 4-bit binary input and outputs its 2-bit code (see the table below for the circuit's truth table). Design the circuit for minimum delay (assuming a stage effort of 4) and driving a load of 10 fF. As part of the design you need to determine the width of all transistors. You can use the following transistor parameters for the design - L = 0.25 micron, minimum transistor width = 0.3 micron, $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, C_g = 1 fF/micron, C_d = 1.5 fF/micron. You can assume that you have both complemented and non-complemented inputs and that input signals have very small rise and fall times. *Hint: Determine the Boolean expression for g_1 and g_0 outputs of the encoder circuit and use the Boolean expression to design the circuit. Use logical effort technique to determine the transistor sizes.* (20 points)

Table 1: Truth table for Encoding

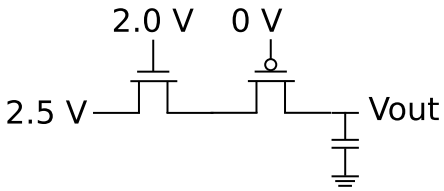
Conventional Binary Encoded Value	New Encoded Value
$b_3\ b_2\ b_1\ b_0$	$g_1\ g_0$
0 0 0 0	0 0
0 0 0 1	0 1
0 0 1 0	1 0
0 0 1 1	1 0
0 1 0 0	0 0
0 1 0 1	0 1
0 1 1 0	1 0
0 1 1 1	1 0
1 0 0 0	1 0
1 0 0 1	1 0
1 0 1 0	1 1
1 0 1 1	1 1
1 1 0 0	0 1
1 1 0 1	0 1
1 1 1 0	0 0
1 1 1 1	0 0

Question 6

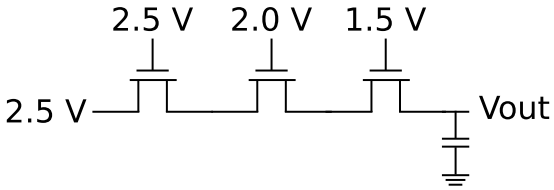
Find the final value of the voltage V_{out} for the following 5 configurations (4 points each). Assume $V_{th,N}$ = 0.5 V and $V_{th,P}$ = -0.5 V. Assume that the capacitor is initially discharged. Ignore subthreshold conduction and body effect.



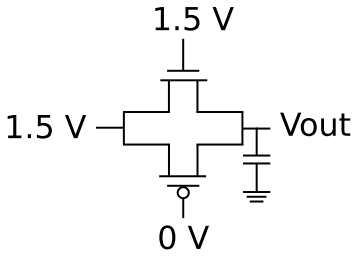
Vout =



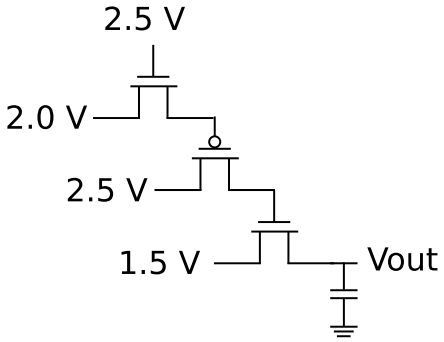
Vout =



Vout =



Vout =



Vout =

Bonus question (5 points)
Prof. Joshi's is a vegetarian and wants to try out some new places in and around campus that serve vegetarian food. Please suggest 5 restaurants/food stalls/food trucks/hole-in-the-wall places that are within walking distance from Photonics Center and may have good vegetarian food options.